Stochastic Processes 2021/22 Semester 1

Tutorial 1

1. Using the moment generating function, find the first and the second moments of a random variable $X \sim \Pi(\lambda)$ having a Poisson distribution with rate $\lambda > 0$.

2. For a random variable $X \sim \Pi(\lambda)$, find $\mathbf{E}(X^2)$ directly from the definition by rewriting the summation as a sum of derivatives.

3. For a random varible $X \sim Exp(\alpha)$ having an exponential distribution with rate $\alpha > 0$ show by induction that
$$\mathbf{E}(X^n) = \frac{n!}{\alpha^n}$$
for all $n \geq 1$.

4. Let $X \sim N(\mu, \sigma)$ be normally distributed with parameters μ and $\sigma > 0$.
   (i) Show that $\mathbf{E}(X) = \mu$ and $\mathbf{Var}\,(X) = \sigma^2$.
   (ii) Let $X \sim N(0, 1)$. Find $\mathbf{E}(X^n)$ for $n = 3, 4, \ldots$.

5. Let a two-dimensional random vector (X, Y) have a uniform distribution in the triangle with vertices $(0, 0)$, $(1, 0)$, and $(0, 3)$, i.e. its density is $f_{X,Y}(x, y) = C$ for points (x, y) from this triangle and $f_{X,Y}(x, y) = 0$, otherwise.
   Find the value of the constant C and the marginal densities $f_X(x)$ and $f_Y(y)$.

6. For the two-dimensional random vector (X, Y) from Question 5, find the expectations $\mathbf{E}(X)$, $\mathbf{E}(Y)$ and $\mathbf{E}(XY)$.

7. Take the density function $f_{X,Y}(x, y)$ from Question 5 and find conditional densities $f_{X|Y}$, $f_{Y|X}$, conditional expectation $\mathbf{E}(X|Y = y)$ and conditional variance $\mathbf{Var}\,(X|Y = y)$.